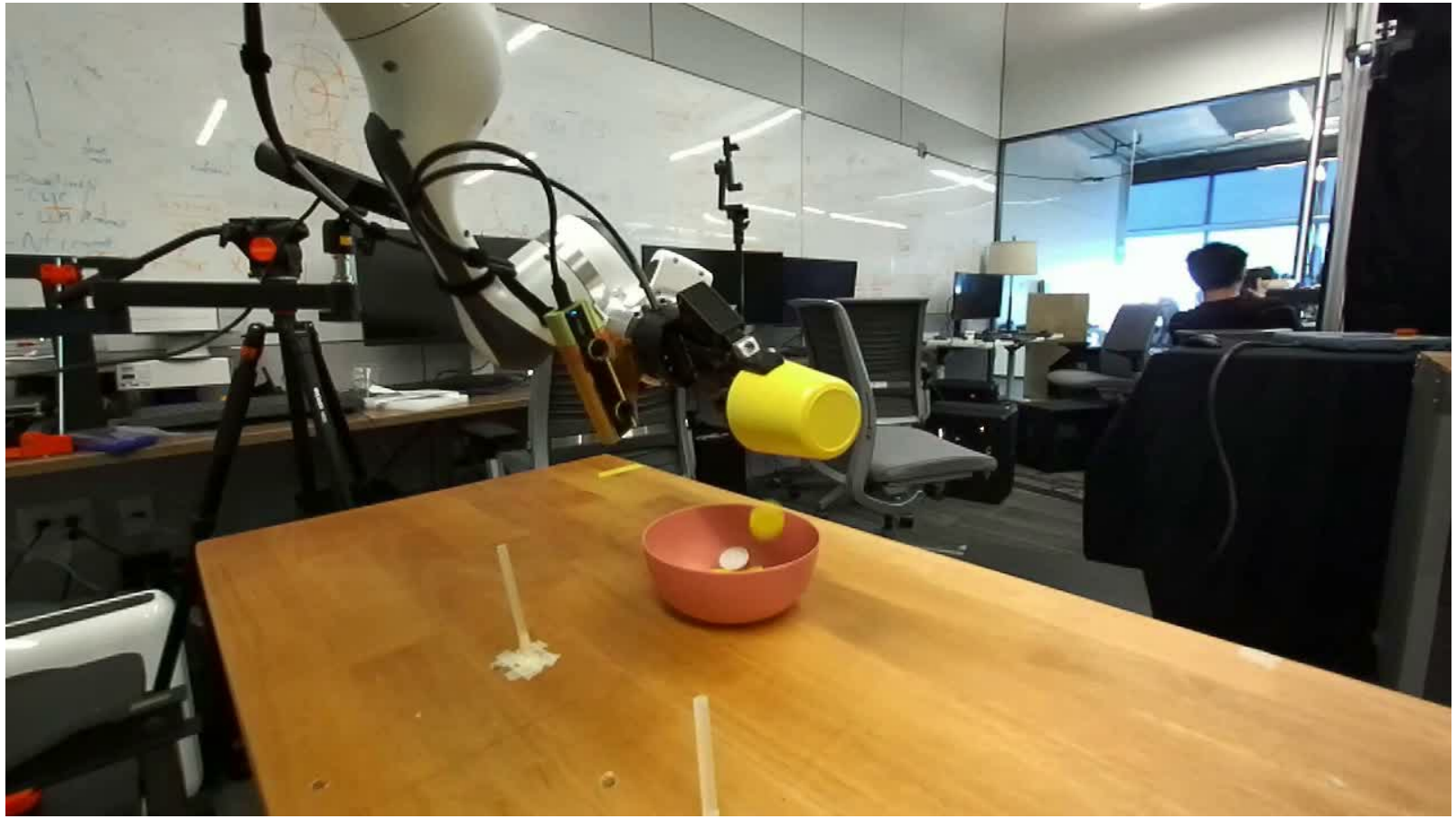
Computational Storage

Aakash, Harshada, Ivan, Jeehan, Mason

Why computational storage?

DROID and robot learning



DROID

76,000

demonstrations

8.7 TB raw

USED IN

π0.5-DROID

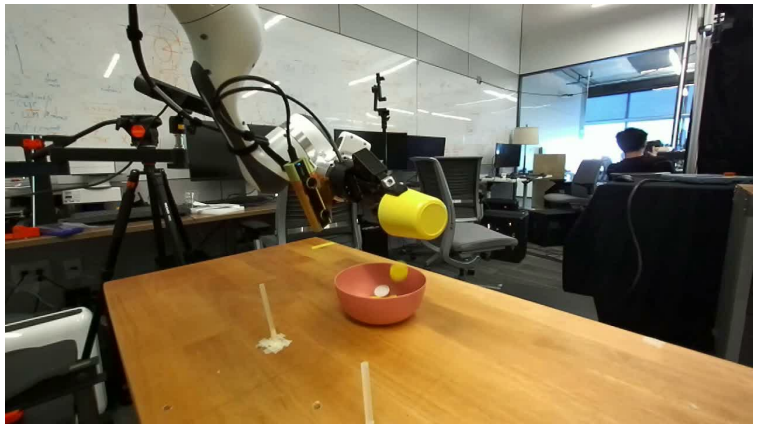
GR00T N1.6

Why computational storage?

Inside one demonstration



EXTERIOR 1



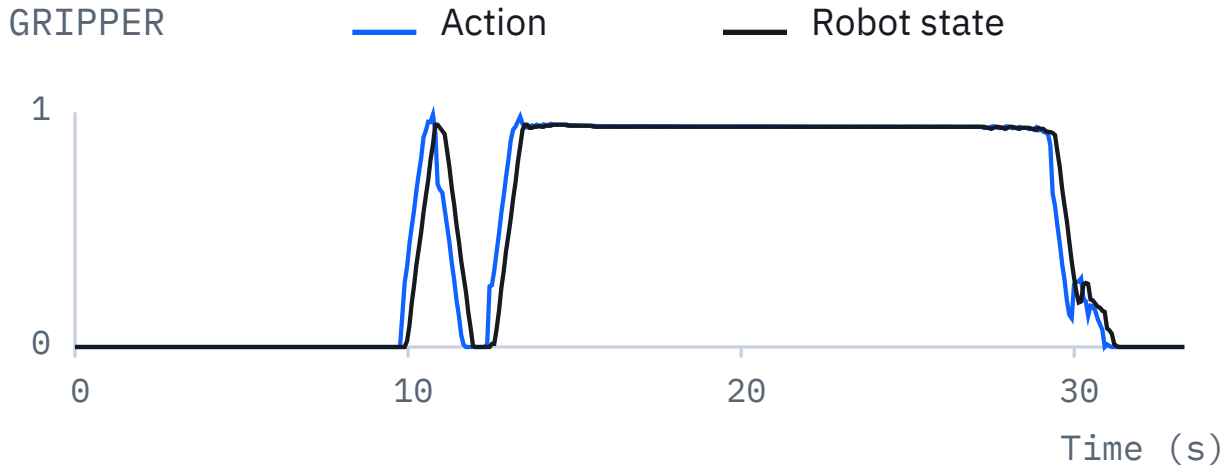
EXTERIOR 2



WRIST

LANGUAGE

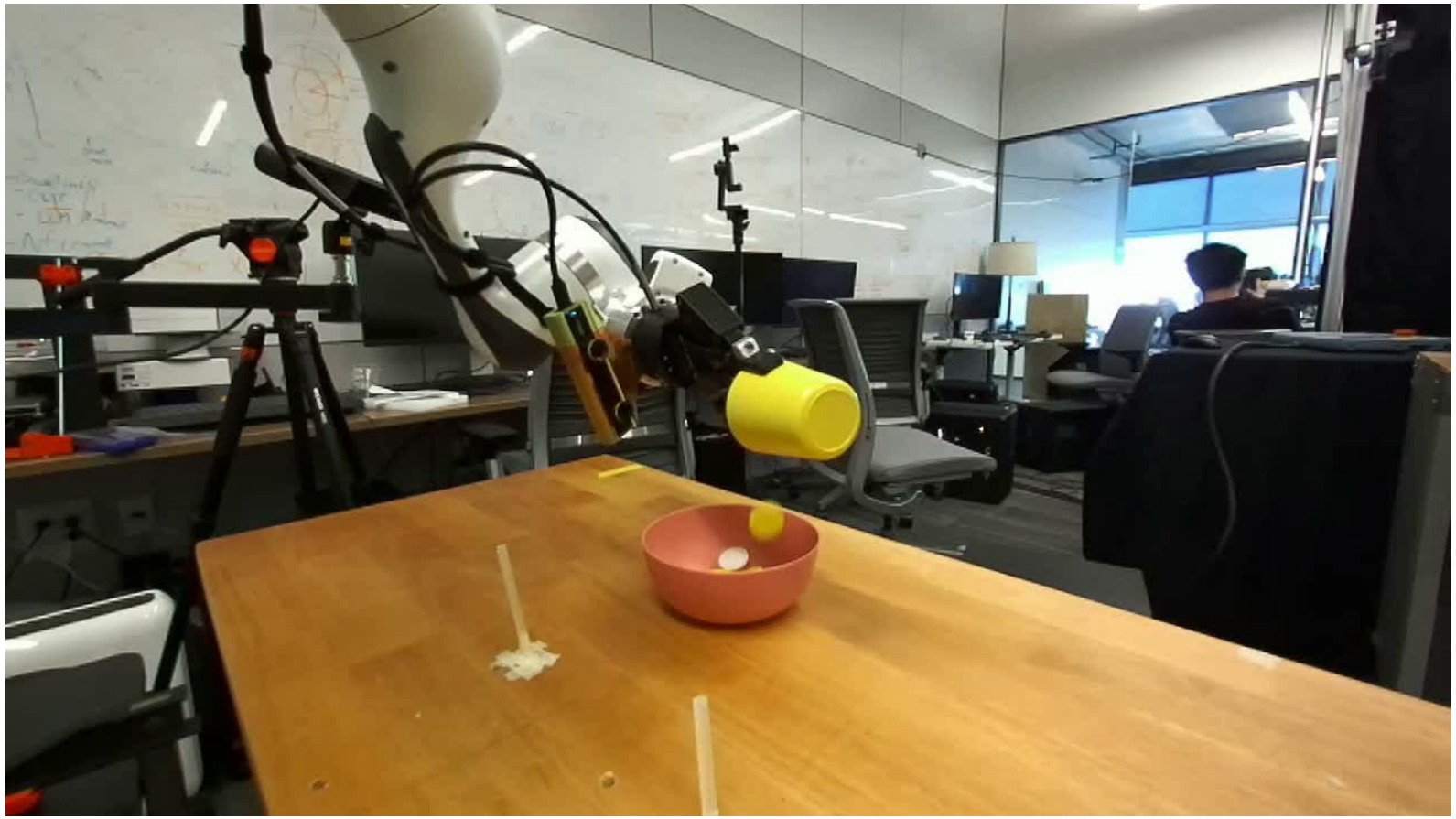
Use a cup to pour.



Why computational storage?

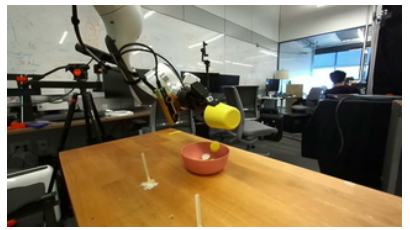
Unnecessary transfers in preprocessing

1280 × 720



16×

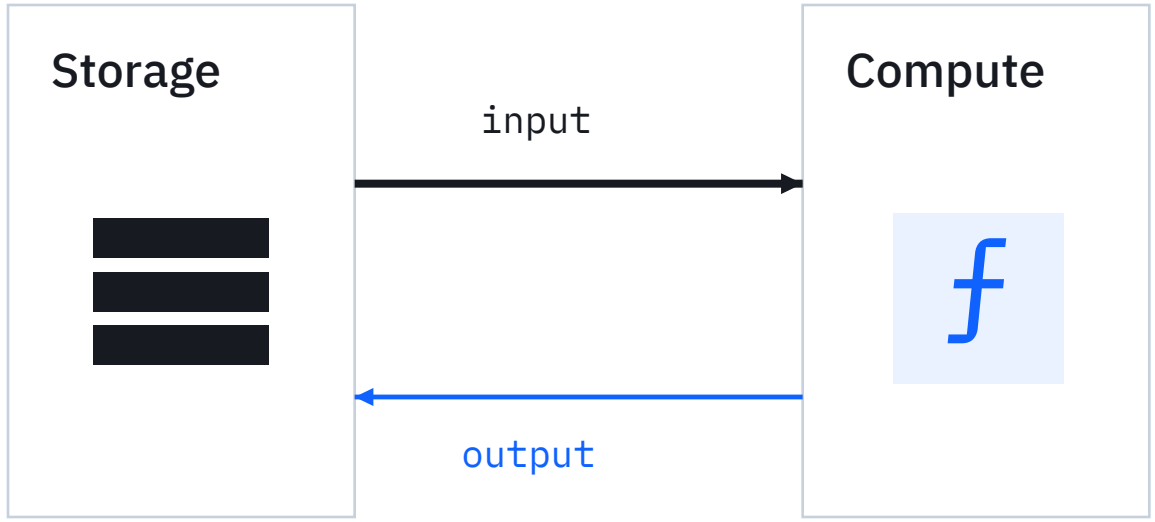
fewer pixels



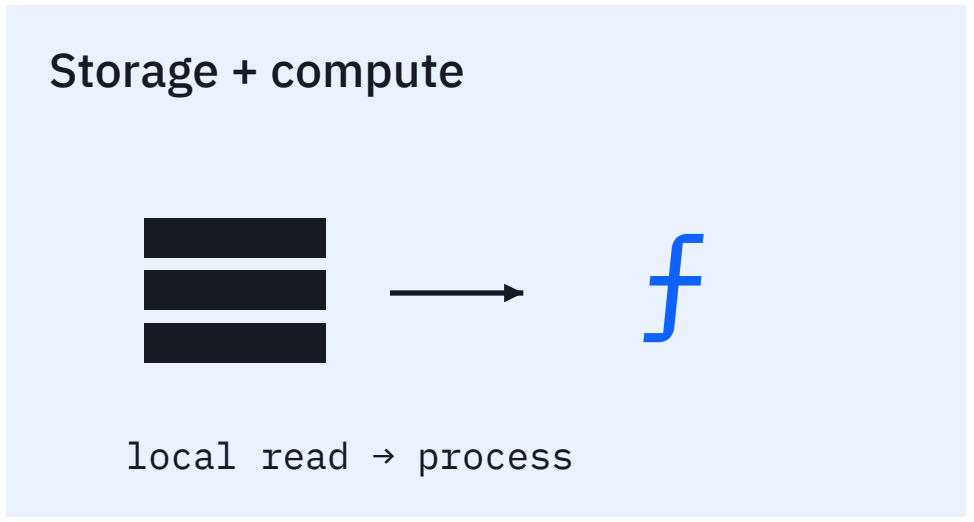
320 × 180

Why computational storage?

Separate

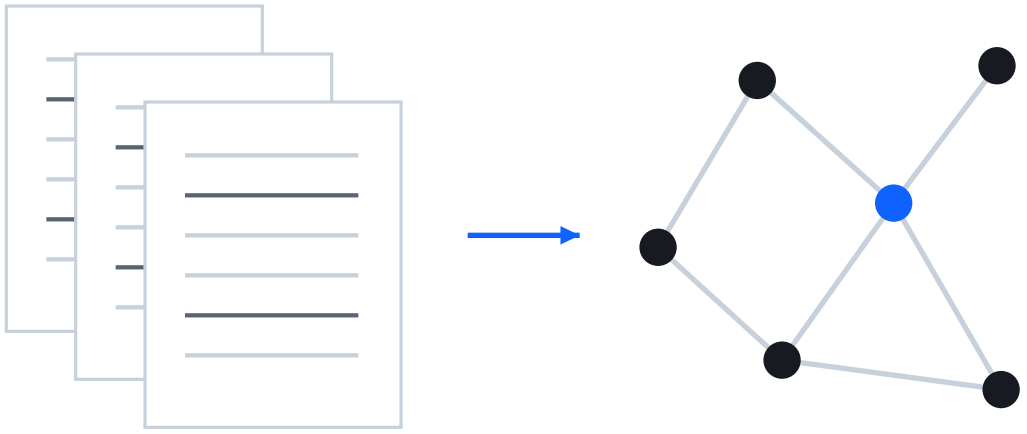


Combined



Why computational storage?

Common Crawl



`web data → link edges`

Argoverse 2 Lidar



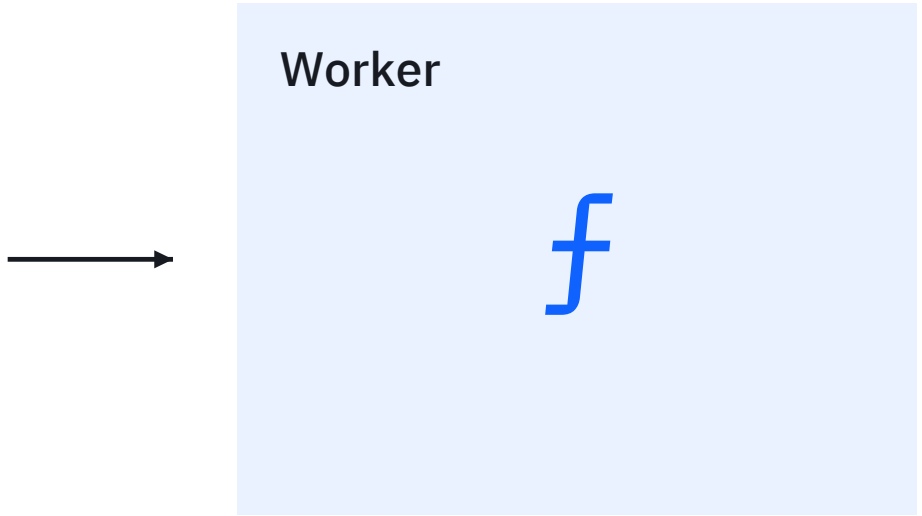
`Point clouds → training windows`

Computational Storage Design

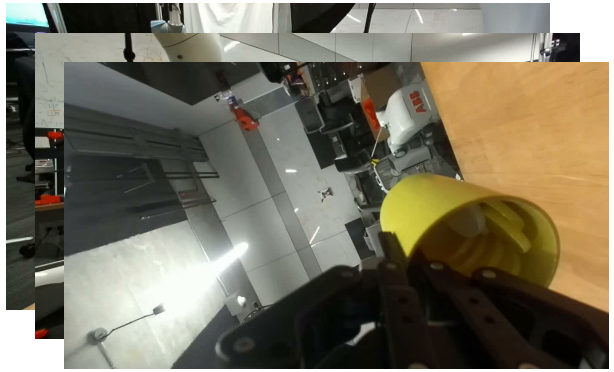
Interface

APPLICATION

`function`	data transformations
`inputs`	domain specific dependencies
`output`	format + destination



Computational Storage Design



video

trajectory.h5

language + metadata

dependency map

Storage node

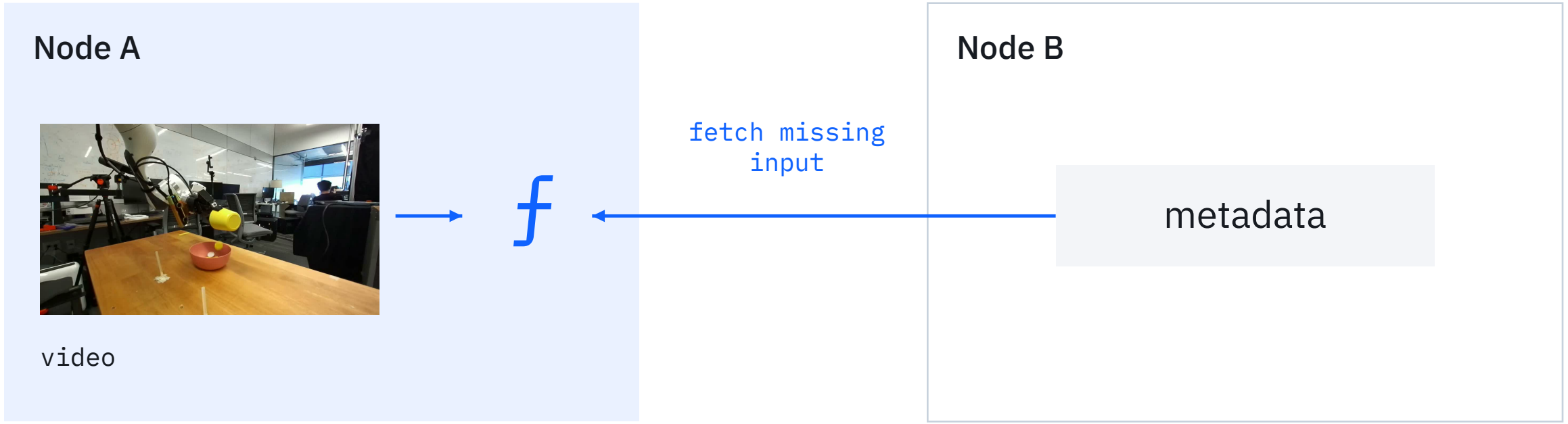
video

trajectory

language

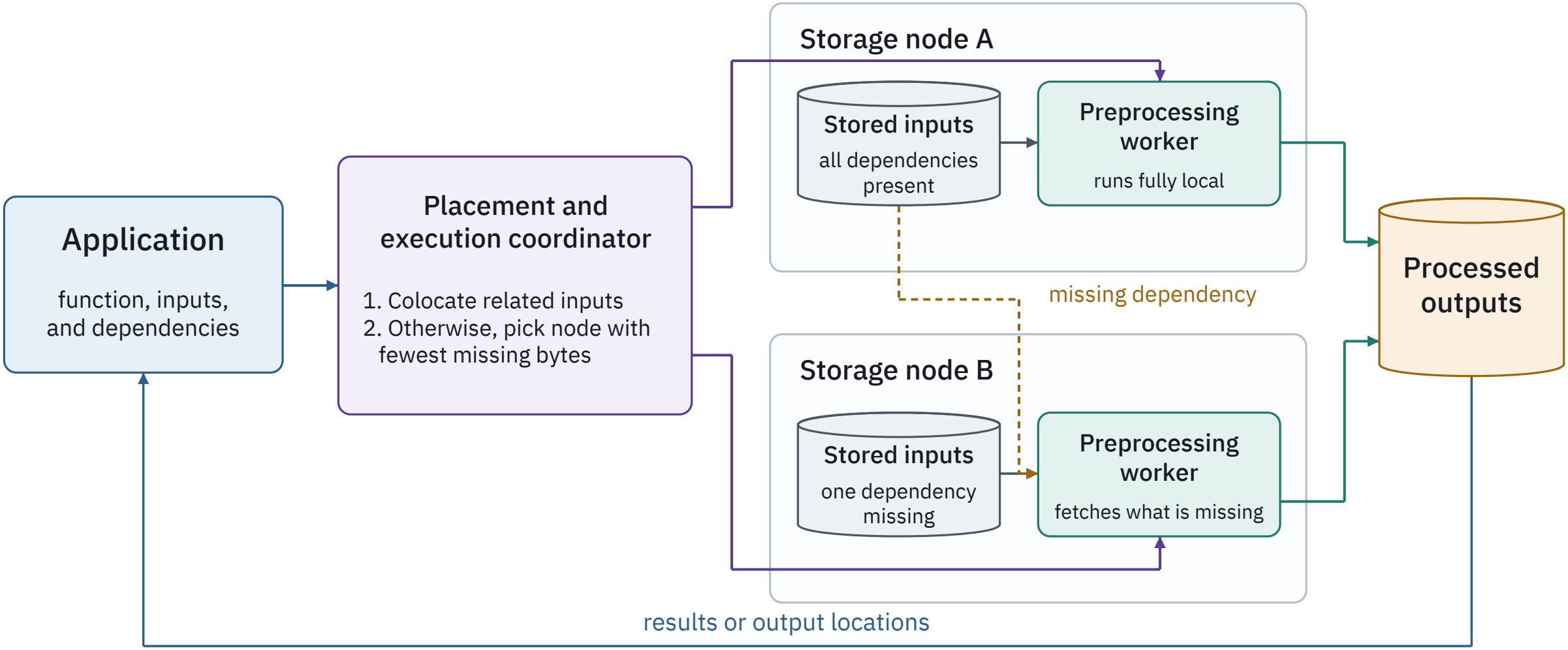
one episode

Computational Storage Design

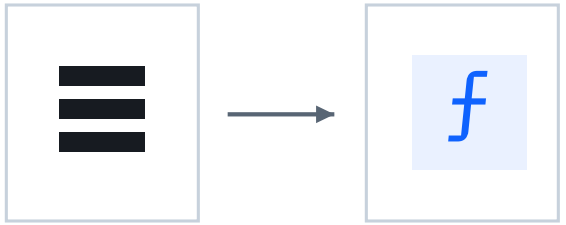
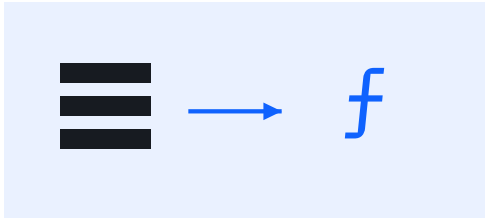
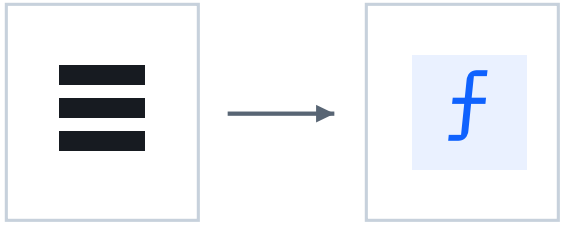
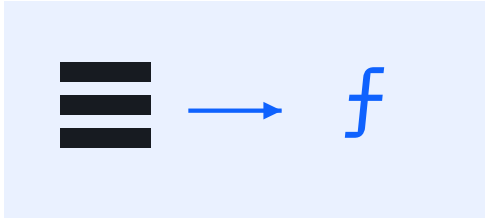


Computational Storage Design

The proposed system



Benchmarks

	Separate	Combined
Raw DROID → RLDS		
RLDS → LeRobot		

Benchmarks

NETWORK BYTES

1.7 TB → 400 GB

≈ −70%

reduction goal

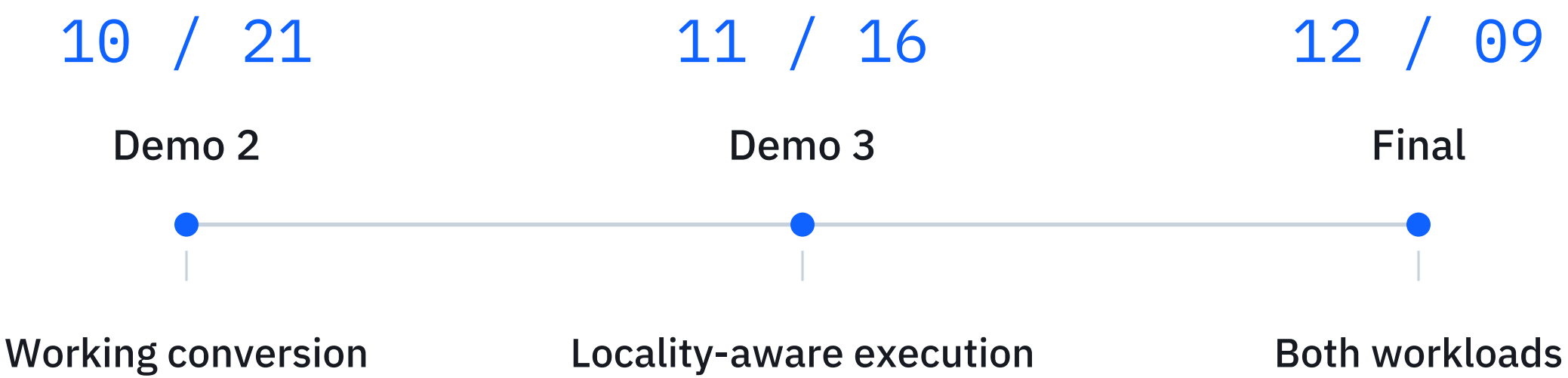
WALL TIME

≤ baseline

OUTPUT

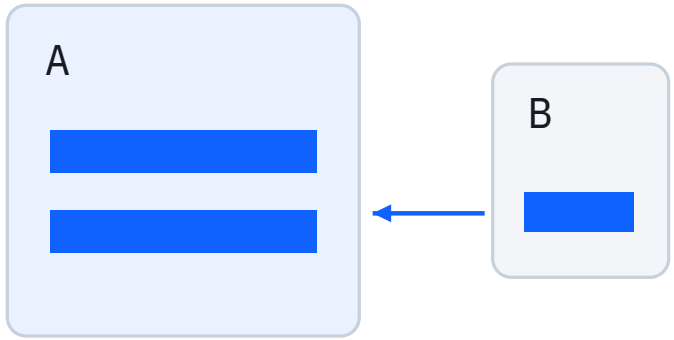
Equivalent

Benchmarks


10 / 21
Demo 2
Working conversion
11 / 16
Demo 3
Locality-aware execution
12 / 09
Final
Both workloads

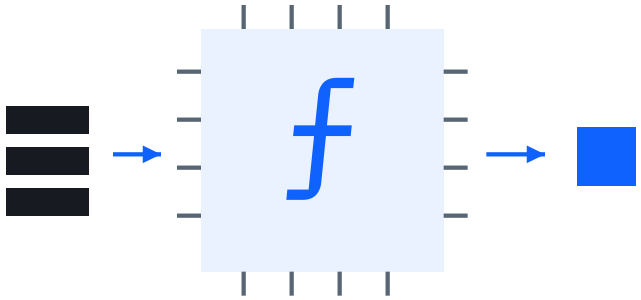
Possible Challenges

PHYSICAL LOCALITY



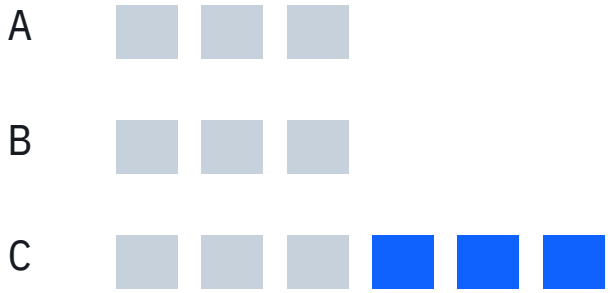
Place dependent data together

ARITHMETIC INTENSITY



Keep operations lightweight

STRAGGLER NODES



Balance work across nodes